

Maharashtra Public Service Commission Assessment

Question Paper Name :	Assistant Professor Mechanical Engineering Group A 15th Dec 2025 Shift 2
Subject Name :	Assistant Professor Mechanical Engineering Group A
Creation Date :	2025-12-15 13:14:50
Duration :	60
Share Answer Key With Delivery Engine :	No
Actual Answer Key :	Yes

Assistant Professor Mechanical Engineering Group A

Group Number :	1
Group Id :	1505616
Group Maximum Duration :	0
Group Minimum Duration :	60
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0

Assistant Professor Mechanical Engineering Group A

Section Id :	1505616
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	100
Number of Questions to be attempted :	100
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	15056111
Question Shuffling Allowed :	Yes

The complementary function of $\frac{d^2y}{dx^2} + a^2y = \sec ax$ is _____ .

Options :

1. $c_1 \cdot \cos ax + c_2 \cdot \sin ax$

2. $(c_1 + c_2) \cdot \cos ax \cdot \sin ax$

3. $c_1 \cdot e^{ax} + c_2 \cdot e^{ax}$

4. $(c_1 + c_2) \cdot e^{ax}$

Laplace transform of a function $f(t)$ is defined for $t > 0$ as _____ .

Options :

1.
$$F(s) = \int_{-\infty}^{+\infty} e^{-st} f(t) dt$$

2.
$$F(s) = \int_{\infty}^0 e^{-st} F(t) dt$$

3.
$$F(s) = \int_0^{\infty} e^{-st} F(t) dt$$

4.
$$F(s) = \int_0^{-\infty} e^{-st} f(-st) dt$$

What is the inverse Laplace transform of $\frac{1}{(s + a)}$?

Options :

1. e^{-at}
2. e^{at}
3. e^{st}
4. e^{sa}

Which among the following method is used to determine numerically largest eigen value and eigen vector of a matrix ?

Options :

1. Power method
2. Liebmann's method
3. Adam's method
4. Euler's method

Which of the following statement is true ?

Options :

1. For a variable x and given sample size, the distribution of variable $\bar{x}$ is called the sampling distribution of the population mean.
2. A normal distribution is determined by the mean and standard deviation.
3. The larger the sample size, the lesser is the standard deviation $\bar{x}$.
4. For sample of size 'n' and the standard deviation of the variable ' $\bar{x}$ ', equals the standard deviations of the variable under consideration multiplied by the square root of the sample size.

What is the main advantage of Runge Kutta method over Euler's method ?

Options :

1. Less computational efforts
2. Better accuracy with similar step size
3. Uses only initial condition
4. No need for function evaluation

What is the probability function of a Poisson distribution ?

Options :

1.
$$f(x) = \binom{n}{x} P^x (1 - P)^{n - x}$$

2.
$$f(x) = \frac{\mu^x}{x!} e^{-\mu}$$

3.
$$f(x) = \frac{x!}{\mu^x} e^{\mu}$$

4.
$$f(x) = e^{-\lambda x}$$

What is an eigen value λ of matrix A ?

Options :

1. A scalar such that $A\vec{x} = \lambda\vec{x}$ has only the zero solution
2. A value for which the equation $A\vec{x} = \lambda\vec{x}$ has a non-zero solution $\vec{x}$
3. A scalar for which the determinant of A is zero
4. A root of polynomial equation $A^2 - \lambda A + I = 0$

Which one of the following is the heat flow equation ?

Options :

1.
$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

2.
$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

3.
$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$$

4.
$$Ax^2 + Bx + c = 0$$

Solve $\frac{d^2y}{dx^2} - 6\frac{dy}{dx} + 9y = 0$.

Options :

1. $y = c_1e^{3x} + c_2e^x$

2. $y = (c_1 + c_2x)e^{3x}$

3. $y = (c_1 + c_2x)e^{2x}$

4. $y = (c_1 + c_2)x$

In differential operator symbol D stands for operation of differential, then $\frac{1}{D^2}$ stands for _____ .

Options :

1. Operation of double derivative
2. Operation of integration
3. Operation of integration twice
4. None of the above

Which one of the following is the Laplace transform of a function $f(t) = e^{50t}$?

Options :

1.
$$\frac{1}{(s - 50)}$$

2.
$$(s - 50)$$

3.
$$\frac{50}{(s^2 - 50)}$$

4.
$$\frac{s^2}{(s^2 + 2500)}$$

Question Number : 13 Question Id : 150561515 Question Type : MCQ Option Shuffling : No

A random variable has a Poisson distribution such that $P(1) = P(12)$. What is the mean distribution ?

Options :

1. 1

2. 3

3. 2

4. 4

Question Number : 14 Question Id : 150561516 Question Type : MCQ Option Shuffling : No

A shelf has 6 mathematics books and 4 physics books. Find the probability that 3 particular mathematics books will be together.

Options :

1. $\frac{1}{15}$

2. $\frac{3}{15}$

3. $\frac{6}{15}$

4. $\frac{8}{15}$

The value of $\lim_{x \rightarrow 0} \frac{x}{\sin x}$ is _____ .

Options :

1. 0
2. 1
3. ∞
4. 2

Which of the following is a correct characteristic of roller bearings compared to ball bearings of comparable size ?

Options :

1. Roller bearings have less fatigue endurance but are cheaper to manufacture.
2. Roller bearings are more flexible and easier to mount.
3. Roller bearings are stiffer and provide greater fatigue endurance.
4. Roller bearings are self-aligning in all applications.

A wedge is generally used for lifting a _____ .

Options :

1. heavy load through large distance
2. heavy load through small distance
3. light load through small distance
4. light load through large distance

Which equation is used to find the angle θ_p at which principal stresses occur ?

Options :

$$\tan 2\theta_p = \frac{\sigma_x - \sigma_y}{2\tau_{xy}}$$

1.

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

2.

$$\tan \theta_p = \frac{\tau_{xy}}{\sigma_x + \sigma_y}$$

3.

$$\tan \theta_p = \frac{\sigma_x + \sigma_y}{\tau_{xy}}$$

4.

The main objective of controlling force is to _____ .

Options :

1. Determine stability of governor
2. Determine sensitiveness of governor
3. Determine both (1) and (2) and determine power of the governor
4. None of the above

Which of the following best describes a system undergoing free vibrations ?

Options :

1. A system oscillating only due to an initial disturbance, with no external force acting afterward.
2. A system oscillating due to an internal heat source.
3. A system oscillating under continuous external force.
4. A system that is motionless due to friction.

In concern to thin cylinders, which of the following statements are correct ?

- a. The thickness of the wall is less than $\frac{1}{15}$ to $\frac{1}{20}$ of its internal diameter.
- b. Thin cylinders are subjected to circumferential and longitudinal stresses.
- c. The circumferential stresses is known as hoops stresses.

Options :

- 1. a and b
- 2. b and c
- 3. a and c
- 4. a, b and c

What is the primary cause of gyroscopic effects in vehicles like motor cycles and aeroplanes ?

Options :

1. Magnetic field
2. Electrical field
3. Rotating masses
4. External torque

Transmissibility is ratio of _____ .

Options :

1. damping force to spring force.
2. force transmitted to the force applied to the machine.
3. force applied to the machine to the force transmitted to the foundation.
4. force transmitted to the foundation to the damping force.

Question Number : 24 Question Id : 150561526 Question Type : MCQ Option Shuffling : No

One way to reduce large acceleration and jerk in a cam mechanism is to _____ .

Options :

1. use smaller cams
2. use larger cams
3. increase the pressure angle
4. reduce the follower mass

Question Number : 25 Question Id : 150561527 Question Type : MCQ Option Shuffling : No

Reverted gear train is used in simple lathes where “back gear” is used to _____ .

Options :

1. give a high speed to the chuck.
2. to lock the tool post.
3. give a slow speed to the chuck.
4. to rotate the chuck in 360° direction.

What is primary characteristics of fatigue failure ?

Options :

1. Material fails at stress higher than its ultimate stress.
2. Material fails suddenly at stress lower than its ultimate tensile strength under fluctuating loads.
3. Material fails due to corrosion.
4. Material fails under constant loads.

The critical speed of rotating shaft depends on _____ .

Options :

1. mass and stiffness
2. mass and eccentricity
3. stiffness and eccentricity
4. mass, stiffness and eccentricity

What is primary reason for considering only half of the torque transmitted when analyzing forces acting on each Kennedy key ?

Options :

1. Because Kennedy key are weaker than shaft.
2. Because torque is equally shared by two keys.
3. Because only one key transmit the full torque.
4. Because keys do not transmit torque directly.

Which of the following statement is true regarding the principle of virtual work in structural analysis ?

Options :

1.

It requires the use of Hooke's Law to determine displacements.

2.

It applies only to systems with linear elastic behaviour.

3.

It requires only equilibrium and compatibility conditions to be satisfied.

4.

It neglects both equilibrium and compatibility conditions.

Which of the following statements are correct with respect to balancing of locomotives ?

- a. Hammer blow is the maximum vertical unbalanced force caused by the mass provided to balance the reciprocating masses.
- b. Hammer blow value is $mr\omega^2$.
- c. It varies as square of the speed.
- d. At high speed it acts vertically upward.

Options :

- 1. a, b and c
- 2. a, c and d
- 3. b, c and d
- 4. a, b, c and d

The overall efficiency η_0 of a power screw is given by which of the following formula ?

Where,

W_ℓ – useful work done

$2\pi M_t$ – is the torque input for one revolution

Options :

1.
$$\eta_0 = \frac{W_\ell}{2\pi.M_t}$$

2.
$$\eta_0 = \frac{M_t}{W_\ell}.2\pi$$

3.
$$\eta_0 = \frac{2\pi W}{M_t.\ell}$$

4.
$$\eta_0 = \frac{W}{2\pi M_t.\ell}$$

Question Number : 32 Question Id : 150561534 Question Type : MCQ Option Shuffling : No

When frictional force helps to apply the brake, the brake is said to be _____ .

Options :

1. Partially self energising
2. Self locking
3. Back stop
4. Self acting

With respect to worm gear, which of the following statements are correct ?

- a. It is used for high speed reduction.
- b. Speed reduction as high as 100 : 1 can be obtained with single pair.
- c. The operation is smooth and silent.
- d. Transmission efficiency is high as compared with other types of gear drives.

Options :

- 1. a and b
- 2. a, b and c
- 3. a, c and d
- 4. a, b, c and d

Which of the following statement best describes the load-sharing-behaviour in a bolted joint when bolt preload is lost ?

Options :

1. All bolts immediately share shear and bearing loads equally.
2. One bolt initially carries the load until yielding distributes shear and bearing among others.
3. Bearing is the primary load-carrying mechanism in a bolted joint even with preload.
4. Clamming friction has no role in bolted joints.

The area of vibration isolation starts when transmissibility is less than unity and $\frac{\omega}{\omega_n}$ is _____ .

(Note : 'ω' is disturbing frequency and ω_n is natural frequency)

Options :

1. < 1
2. $= 1$
3. $> \sqrt{2}$
4. Zero

Question Number : 36 Question Id : 150561538 Question Type : MCQ Option Shuffling : No

Select incorrect type of frictional clutch.

Options :

1. Disc or plate clutch
2. Cone clutch
3. Centrifugal clutch
4. None

In thrust bearings, the load acts _____ .

Options :

1. along the axis of rotation.
2. parallel to the axis of rotation.
3. perpendicular to the axis of rotation.
4. in any direction.

In which type of machines are flywheels most commonly used ?

Options :

1. Machines that operate in random motion
2. Machines that run on hydraulic power
3. Machines with no rotational parts
4. Machines operating in steady-state conditions

The velocity ratio of a differential wheel and axle with 'D' as diameter of effort wheel and 'd₁' and 'd₂' as diameters of larger and smaller axles respectively is _____ .

Options :

1.
$$\frac{2D}{d_1 + d_2}$$

2.
$$\frac{2D}{d_1 - d_2}$$

3.
$$\frac{D}{d_1 + d_2}$$

4.
$$\frac{D}{d_1 - d_2}$$

Two shafts A and B are made of the same material. The diameter of the shaft A is twice as that of shaft B. The power transmitted by the shaft A will be _____ of shaft B.

Options :

1. twice
2. four times
3. eight times
4. sixteen times

Which law states that the shear stress (τ) on a fluid element layer is directly proportional to the rate of shear strain ?

Options :

1. Newton's law of viscosity
2. Newton's law of specific viscosity
3. Pascal's law
4. Euler's model law

Gauge pressure at a point is equal to _____ .

Options :

1. absolute pressure plus atmospheric pressure
2. absolute pressure minus atmospheric pressure
3. vacuum pressure plus absolute pressure
4. absolute pressure minus vacuum pressure

The total pressure on a plane surface immersed in a static mass of fluid acts in the direction _____ .

Options :

1. parallel to the plane surface
2. normal to the plane surface
3. inclined to the plane surface
4. away from the plane surface

While determining a π (pi) term using Buckingham's π theorem, if, m is number of variables and n is number of fundamental dimensions, the number of π term is one (1) then, which of the following is correct ?

Options :

1. $m < n$
2. $m = n$
3. $m = n - 1$
4. $m = n + 1$

Which of the following statement is correct with respect to incompressible flow ?

- a. It is that of flow in which the density is constant for the fluid flow.
- b. Liquids are incompressible.
- c. Gases are compressible.
- d. Mathematically, density $\rho = \text{constant}$.

Options :

1. a, c and d

2. a, b and d

3. b, c and d

4. a and d

Loss of head due to bend in pipe is calculated by _____ .

Options :

1.
$$\frac{kV^2}{2g}$$

2.
$$\frac{kV}{g}$$

3.
$$\frac{kV}{2g^2}$$

4.
$$\frac{2kV}{g}$$

Which among the following statements is correct for Bernoulli's equation ?

Options :

1. If the flow is irrotational then the same Bernoulli's equation is applicable to all the points in the flow field.
2. For rotational flow the Bernoulli's equation is applicable for a particular streamline.
3. Both (1) and (2)
4. None of the above

In case of balanced counter flow heat exchangers, the effectiveness is measured by equation _____ .

Options :

1.
$$\varepsilon = \frac{NTU}{1 - NTU}$$

2.
$$\varepsilon = \frac{NTU}{1 + (NTU \times LMTD)}$$

3.
$$\varepsilon = \frac{NTU}{1 + LMTD}$$

4.
$$\varepsilon = \frac{NTU}{1 + NTU}$$

Question Number : 49 Question Id : 150561551 Question Type : MCQ Option Shuffling : No

Out of the following which factor will not affect thermal conductivity of the insulators ?

Options :

1. Temperature

2. Density

3. Air speed

4. Moisture

The temperature variation with time, in the lumped parameter model is _____ .

Options :

1. Exponential
2. Cubic
3. Logarithmic
4. Linear

The lumped capacitance method can be used for which of the following conditions ?

Options :

1. $B_i = \frac{hL_c}{k} < 0.1$

2. $B_i = \frac{hL_c}{k} = 0.1$

3. $B_i = \frac{hL_c}{k} > 0.1$

4. All of the above

If pressure tends to zero with ideal gas compressibility factor will tends to _____ .

Options :

1. Zero

2. Unity

3. More than 3

4. Inbetween 2 to 3

In Carnot cycle which of the following assumption is incorrect ?

- a. The piston moving in a cylinder does not develop any friction during motion.
- b. The transfer of heat loss not affect the temperature of source or sink.
- c. Working medium is perfect gas and has constant specific heat.
- d. Compression and expansion are irreversible.

Options :

- 1. a
- 2. c
- 3. d
- 4. b

Nusselt number is given by _____ .

Options :

1.
$$\frac{\text{Internal thermal resistance}}{\text{Surface resistance}}$$

2.
$$\frac{\text{Momentum diffusivity}}{\text{Thermal diffusivity}}$$

3.
$$\frac{\text{Buoyancy force}}{\text{Viscous force}}$$

4.
$$\frac{\text{Inertia force}}{\text{Viscous force}}$$

A system which consists of single phase is termed as _____ .

Options :

1. Open system
2. Closed system
3. Heterogeneous system
4. None of the above

In case of reciprocating compressor, volumetric efficiency is defined as _____ .

Options :

1. Actual volume of gas drawn at evaporator pressure and temperature/piston displacement.
2. Actual volume of gas drawn at condenser pressure and temperature/piston displacement.
3. Actual volume of gas drawn at atmospheric pressure and temperature/piston displacement.
4. Actual volume of gas drawn at evaporator pressure and temperature/speed of piston.

What is the primary purpose of steam bleeding (regeneration) in a steam power plant ?

Options :

1. To increase the moisture content in the turbine.
2. To improve the thermal efficiency of the cycle.
3. To reduce the feed water temperature.
4. To increase the back pressure in the condenser.

Which of the following statement is true ?

Options :

1. Air is not a pure substance but liquid air is a pure substance.
2. Air is a pure substance but liquid air is not a pure substance.
3. The mixture of air and liquid air is a pure substance.
4. The mixture of air and liquid air is not a pure substance.

The ratios of pressure, temperature and specific volume of a real gas to the corresponding critical values are called _____ .

Options :

1. Reduced properties
2. Extensive properties
3. Intensive properties
4. None of the above

A gas compressor is a device in which work is done on the gas to raise its _____ with appreciable increase in its _____ .

Options :

1. volume, temperature
2. temperature, pressure
3. pressure, density
4. volume, density

In 4-stroke petrol engine, the angle through which the spark is given earlier is known as “Ignition Advance” or “Angle of Advance”. This varies from _____ rotation of crank as per the speed of the engine.

Options :

1. 10° to 20°
2. 20° to 50°
3. 30° to 40°
4. 20° to 40°

Which cycle is termed as limited pressure cycle ?

Options :

1. Otto cycle
2. Diesel cycle
3. Dual cycle
4. Brayton cycle

A Carnot engine working between 427°C and 77°C produces 130 kJ of work.

What is the engine thermal efficiency ?

Options :

1. 52%
2. 50%
3. 51%
4. 63%

Sigma heat function is _____ .

Options :

1. Constant along the constant DBT line.
2. Constant along the constant DBT and WBT line simultaneously.
3. Constant along the constant enthalpy line.
4. Constant along the constant WBT line.

With reference to unit of refrigeration which of the statement is incorrect ?

Options :

1. Unit of refrigeration is tonne of refrigeration.
2. It is a amount of refrigeration effect produced by one tonne of ice from and at -4°C in 12 hours.
3. $1 \text{ TR} = 210 \text{ kJ/min.}$
4. $1 \text{ TR} = 3.5 \text{ kW.}$

The relation between hydraulic efficiency (η_h), mechanical efficiency (η_m) and overall efficiency (η_o) is given by _____ .

Options :

1. $\eta_h = \eta_m \times \eta_o$

2. $\eta_o = \eta_m \times \eta_h$

3. $\eta_o = \frac{\eta_m}{\eta_h}$

4. $\eta_h = \frac{\eta_m}{\eta_o}$

What is sensible heat factor ?

Options :

1. Ratio of sensible heat transfer to latent heat transfer
2. Ratio of total heat transfer to sensible heat transfer
3. Ratio of sensible heat transfer to total heat transfer
4. Ratio of latent heat transfer to sensible heat transfer

What is the disadvantage of vapour compression refrigeration system over air refrigeration system ?

Options :

1. It has smaller size for given capacity of refrigeration
2. It has less running cost
3. The initial cost is high
4. It can be employed over a large range of temperature

The gas turbine works on _____ .

Options :

1. Carnot cycle
2. Stirling cycle
3. Brayton cycle
4. None of the above

Operating characteristic curves of a turbine means _____ .

Options :

1. Curves drawn at constant head and speed
2. Curves drawn at constant head
3. Curves drawn at constant speed
4. Curves drawn at constant efficiency

Which are the constituent metals of an alloy stainless steel ?

Options :

1. Copper and Zinc
2. Iron, Nickel and Chromium
3. Copper and Tin
4. Mercury and Iron

With increasing temperature, values of elastic modulus _____, tensile and yield strength _____, ductility _____.

Options :

1. Increase, Increase, Increase
2. Decrease, Decrease, Decrease
3. Increase, Increase, Decrease
4. Decrease, Decrease, Increase

Which of the following statements is true about endurance limit ?

Options :

1. It is the maximum stress a material can withstand in a single force application.
2. It is the stress level below which a material can withstand infinite loading cycles without failure.
3. It is the temperature limit for a material under creep loading.
4. It is the average stress level during fatigue.

A fusion weld between two dissimilar metals can be possible only when _____ .

Options :

1. They have same melting points
2. They have metallurgical solubility
3. They have identical thermal conductivity
4. Both are non ferrous metals

Which feature differentiates soldering from brazing in stainless steel applications ?

Options :

1. Soldering creates stronger joints than brazing
2. Soldering eliminates the need to remove chromium oxide films
3. Soldering generally uses lower temperature and tin based alloys
4. Soldering is always preferred for high strength applications

Read the following statements related to weldability.

- a. Metal composition plays a dominant role in determining the weldability.
- b. Plain carbon steels containing more than 0.25% carbon have best weldability.
- c. The weldability can be judged from the carbon equivalent of the steel.
- d. Hardenability and weldability properties are equivalent to each other.

Which of the above statement is/are not correct ?

Options :

- 1. a and c
- 2. b and d
- 3. c only
- 4. d only

Question Number : 77 Question Id : 150561579 Question Type : MCQ Option Shuffling : No

A gating system having a sprue of 1 cm^2 , a runner of 3 cm^2 and three gates, each having 1 cm^2 cross-sectional area. What is the gating ratio ?

Options :

1. $1 : 1 : 3$

2. $3 : 1 : 3$

3. $3 : 3 : 1$

4. $1 : 3 : 3$

In deep drawing, a flat metal sheet is converted into a _____ .

Options :

1. Cylindrical rod
2. Hollow pipe
3. Cup shaped articles
4. Forged components

Which of the following forging defect is caused due to improper forging and faulty die design ?

Options :

1. Scale pits
2. Cracks due to overheating
3. Laps at corners or a surfaces
4. Fibre flow disruption

What does the acronym LASER stand for ?

Options :

1. Light Amplification by Spontaneous Emission of Radiation
2. Light Amplification by Simulated Emission of Radiation
3. Light Amplification by Stimulated Emission of Radiation
4. Light Adjustment by Stimulated Energy of Reflection

Machining allowance in casting process is provided on a pattern to _____ .

Options :

1. make it easier to withdraw from the mold.
2. allow post casting material removal.
3. avoid sharp corners.
4. increase the weight of casting.

In medium alloy tool steels chromium and molybdenum are added to increase _____ and tungsten is added to increase _____.

Options :

1. wear resistance, hardenability
2. hardenability, wear resistance
3. grain size, wear resistance
4. wear resistance, grain size

Which of the following is not the difference between capstan and turret lathe ?

Options :

1. The turret of the capstan lathe is mounted on a short slide and the turret of the turret lathe is mounted on a saddle.
2. Larger and heavier chucking works are usually handled on turret lathe, whereas a capstan lathe is suitable for bar work.
3. On the capstan lathe, turret can be moved rapidly, whereas it is laborious in turret lathe.
4. None of the above

Which among the following is true for cutting cost ?

Options :

1. Cutting cost per piece depends on the time the tool is actually cutting
2. In proportion to total cost, cutting cost is small in small job shops
3. Both (1) and (2) are not true
4. Both (1) and (2) are true

In turning operation of machining, feed is defined as _____ .

Options :

1. Movement of tool per second
2. Movement of workpiece per revolution
3. Lateral movement of tool per revolution of the workpiece
4. Movement of the chuck jaws

Which of the following expression is used to determine the safety stock ?

Options :

$$\text{Safety stock} = \frac{(\text{Maximum lead time} - \text{Normal lead time})}{\text{Consumption rate}}$$

1.

$$\text{Safety stock} = (\text{Normal lead time} - \text{Max. lead time}) \times \text{Consumption rate}$$

2.

$$\text{Safety stock} = (\text{Max. lead time} - \text{Normal lead time}) \times \text{Consumption rate}$$

3.

None of the above

4.

Which of the following is incorrect in the context of limits and fits ?

Options :

1. $\text{Upper deviation} = \text{Max. limit of size} - \text{Basic size}$

2. $\text{Lower deviation} = \text{Min. limit of size} - \text{Basic size}$

3. $\text{Tolerance} = \text{Max. limit of size} - \text{Minimum limit of size}$

4. $\text{Basic size} = \frac{\text{Max. limit} - \text{Min. limit}}{2}$

Under which category does an autocollimator fall, considering its primary function ?

Options :

1. Tool for measuring straight line distances
2. Equipment for shaping cylindrical components
3. Device for precise optical angle measurement
4. Instrument for evaluating circularity

Read the following statements.

- a. Investing in new types of machines to manufacture new product designs.
- b. Material requirement planning.
- c. Purchase of existing plants from other companies.
- d. New plant construction.

Which of the above are not included as the decisions required for capacity planning ?

Options :

- 1. a and b
- 2. c and d
- 3. b only
- 4. c only

Which of the following system denotes a user-oriented system in which the computer is employed to create, transform and display data in the form of pictures or symbols ?

Options :

1. Computer Process Control (CPC)
2. Computer Aided Process Planning (CAPP)
3. Direct Numerical Control (DNC) System
4. Interactive Computer Graphics (ICG)

Group Technology in CAD/CAM is most likely used for the purpose to _____ .

Options :

1. reduce inventory
2. increase drawing speed with accuracy
3. classify and code similar parts for process planning
4. isolate data storage for security

Read the following statements related to methods to solve the linear programming problem

- a. The simplex method is used when we have two decision variables in the problem.
- b. In graphical method, the problem may have any number of decision variables.
- c. In graphical method, the inequalities are assumed to be equations.
- d. In simplex method, the presence of slack variable indicates the idle capacity of the resources.

Options :

- 1. b only
- 2. a and b
- 3. c only
- 4. c and d

Question Number : 93 Question Id : 150561595 Question Type : MCQ Option Shuffling : No

If a network is constructed on the basis of jobs or activities it is called as _____ .

Options :

1. JIT
2. FIFO
3. PERT
4. CPM

Which of the following methods is not the method for basic solution of the transportation model ?

Options :

1. North West Corner Method
2. Row Minima Method
3. Column Minima Method
4. The Hungarian Method

Which of the following environments typically has many raw material components but relatively few final products ?

Options :

1. Make-to-stock (MTS)
2. Assemble-to-order (ATO)
3. Make-to-order (MTO)
4. Engineer-to-order (ETO)

The two basic types of forecasting are _____ .

Options :

1. Deterministic and probabilistic
2. Deterministic and non deterministic
3. Time series and quantitative
4. Qualitative and quantitative

What is the purpose of value stream mapping in lean production ?

Options :

1. To track customer orders in real-time
2. To analyze financial performance
3. To map and analyze all activities inventories and information flows
4. The forecast demand fluctuations

In JIT system, what can be done with excess capacity when demand is below production capability ?

Options :

1. Increase production to store inventory
2. Send workers home
3. Use time for non-production activities like training and maintenance
4. Fire employees to reduce costs

Which type of inventory helps different production operations work independently despite having different capacities ?

Options :

1. Buffer inventory
2. Cycle inventory
3. Anticipation inventory
4. Decoupling inventory

Obsolescence and spoilage are both examples of _____ .

Options :

1. Cost of not having inventory
2. Cost of having inventory
3. Ordering costs
4. Stock out costs